

DIABETES DATA ANALYSIS

QUESTION 1.1: CORRELATION MATRIX AND VARIABLE RELATIONSHIPS

A correlation matrix helps us understand **how strongly variables move together**. In the diabetes dataset, several important patterns emerge.

I. VERY STRONG CORRELATIONS ($|r| > 0.85$)

S1 and S2 ($r = 0.90$)

These two serum measurements move almost exactly together.

- They contain nearly the same information
- This is a clear case of **severe multicollinearity**
- Including both in a regression model is unnecessary
- One of them should normally be removed to avoid problems

Key idea: When two predictors are this similar, the model struggles to tell their individual effects apart.

II. STRONG CORRELATIONS ($0.60 < |r| < 0.85$)

Several serum variables are still strongly related, though not identical:

- **S2 $\leftrightarrow$ S4 ($r = 0.66$)**
- **S4 $\leftrightarrow$ S5 ($r = 0.62$)**
- **S3 $\leftrightarrow$ S4 ($r = -0.74$)** (strong negative relationship)

Interpretation:

- Many serum measurements form **connected biological groups**
- S3 behaves in the opposite direction compared to S4 and S5
- This suggests **different physiological processes** acting against each other
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III. MODERATE CORRELATIONS ($0.30 < |r| < 0.60$)

Body Mass Index (BMI):

- Moderately related to S4, S5, and S6
- Indicates body composition affects blood chemistry

Blood Pressure (BP):

- Moderately related to BMI, S5, and S6
- Shows BP is influenced by both body structure and serum factors

Age:

- Most strongly linked to BP ($r = 0.34$)
- Weakly related to most other variables

Overall interpretation:

Physical characteristics like BMI and BP act as **bridges** between demographic variables and serum measurements.

IV. WEAK CORRELATIONS ($|r| < 0.30$)

- **Age** has little connection with most serum variables
- **Sex** shows very weak relationships overall

These variables provide **relatively independent information**, which can be useful in modeling.

V. KEY PATTERNS IDENTIFIED

1. **Serum Cluster:** S1, S2, S4, and S5 are positively correlated

2. **Opposition Effect:** S3 moves in the opposite direction of S4 and S5
3. **BMI as a Connector:** Links body characteristics to serum chemistry
4. **Relative Independence:** Age and Sex stand mostly alone
- 5.

VI. MODELING IMPLICATIONS

- Multicollinearity is expected in regression models
- Variable selection becomes essential
- The S1–S2 pair needs special handling
- Some predictors may appear insignificant even when they are important
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QUESTION 1.2: COLLINEARITY – MEANING AND EFFECTS

I. WHAT IS COLLINEARITY?

Collinearity occurs when **predictor variables are strongly correlated**, meaning one can be predicted from others.

Types of collinearity:

- **Perfect:** Exact relationship (e.g., $X_2 = 3X_1 + 5$)
- **High:** Strong but not exact (correlation > 0.8)

How we detect it:

- High correlations in the matrix
- Variance Inflation Factor (VIF) greater than 10
- Poorly conditioned $X'X$ matrix

II. EFFECTS ON REGRESSION COEFFICIENTS

A. Variance Inflation

- Standard errors become very large
- Confidence intervals widen
- t-statistics shrink
- Truly important variables may appear insignificant

As predictors become more correlated, the variance of coefficient estimates increases dramatically.

B. Instability of Estimates

- Small data changes cause large coefficient changes
- Signs of coefficients may flip unexpectedly
- Coefficient size becomes unreliable

C. Loss of Statistical Power

- High chance of **Type II error** (missing real effects)
- Individual t-tests lose meaning
- Overall model may still look good (high R^2)

D. Interpretation Problems

- Impossible to isolate the effect of a single variable
- “Holding other variables constant” becomes unrealistic
- Coefficients lose practical interpretability

IV. EXAMPLE FROM THE DIABETES DATA

S1 and S2 ($r = 0.90$):

- Share about **81% of the same information**

- Their individual effects cannot be separated
- Both get inflated standard errors
- One may appear insignificant even though both matter together

V. SOLUTIONS TO COLLINEARITY

- Remove one variable from correlated pairs
- Use ridge regression
- Apply Principal Component Analysis (PCA)
- Perform variable selection
- Use medical and domain knowledge

QUESTION 1.3: IS COLLINEARITY A PROBLEM HERE?

A. ARE ALL VARIABLES SIGNIFICANT?

Usually, **no**.

Some variables appear insignificant even though theory suggests they matter. This mismatch is a warning sign.

B. IS THIS A COLLINEARITY PROBLEM?

Yes, very clearly.

Evidence:

1. **From correlations:** Extremely high links among serum variables
2. **From theory:** Biological measurements are naturally related
3. **From results:** High R^2 but weak individual t-tests
4. **From mathematics:** Inflated variances and unstable estimates
- 5.

C. CONSEQUENCES FOR THE MODEL

- Individual coefficients are unreliable
- Significance tests are misleading
- Overall model fit is acceptable
- Predictions are still usable
- Interpretation is difficult

D. RECOMMENDED ACTIONS

1. Calculate VIF values
2. Apply variable selection methods
3. Remove one of S1 or S2
4. Consider ridge regression
5. Use domain knowledge

E. CONCLUSION

The combination of:

- High correlations
- Weak individual significance
- Good overall fit

Strongly confirms that collinearity is distorting the regression results.

QUESTION 1.4: FORWARD VS BACKWARD VARIABLE SELECTION

I. FORWARD SELECTION

Idea: Start simple and add variables gradually.

- Begin with no predictors
- Add the most significant variable at each step
- Stop when no remaining variable is significant

Strengths:

- Fast and efficient
- Works when predictors exceed observations
- Handles collinearity better than backward

Weaknesses:

- Early mistakes cannot be corrected
- May miss suppressor effects
- Not guaranteed optimal

II. BACKWARD SELECTION

Idea: Start with everything and remove weak predictors.

- Begin with all variables
- Remove the least significant one at each step
- Stop when all remaining predictors are significant

Strengths:

- Considers full context initially
- Detects suppressor variables
- Better theoretical coverage

Weaknesses:

- Requires many observations
- Computationally expensive
- Can struggle with strong collinearity

III. COMPARISON SUMMARY

Forward builds up; backward trims down.

Each has strengths depending on **sample size, collinearity, and goals.**

QUESTION 1.5: STEPWISE SELECTION

Stepwise selection combines the ideas of forward and backward methods.

WHY STEPWISE IS USEFUL HERE

- Automatically handles S1–S2 redundancy
- Chooses the strongest representative variable
- Produces simpler, interpretable models
- Reduces overfitting
- Improves generalization

LIMITATIONS TO REMEMBER

- Greedy, not optimal
- Sensitive to sample changes
- Multiple testing issues

- Ignores interaction effects unless added manually

APPLICATION TO DIABETES DATA

- Confirms collinearity concerns
- Selects fewer but meaningful predictors
- Achieves similar performance with simpler models
- Improves interpretability and stability

QUESTION 2: TITANIC DATA ANALYSIS

QUESTION 2.1: LOGISTIC REGRESSION VS LINEAR REGRESSION

I. BASIC DIFFERENCES

A. Type of Output (Dependent Variable)

Linear Regression

Linear regression is used when we want to predict a **continuous value** — something that can take many numerical values.

For example, predicting:

- House prices
- Temperature
- Test scores

The output can be any real number, whether positive or negative.

Logistic Regression

Logistic regression is used when the outcome is **binary or categorical**, meaning there are only two possible results.

Examples include:

- Survived or died
- Pass or fail
- Yes or no

Here, the output is limited to **0 or 1**, representing two classes.

B. Nature of Predictions

Linear Regression

Because of this, predictions can go beyond logical limits. For example, it might predict a survival value of 1.3, which makes no sense.

Logistic Regression

Instead of predicting a raw value, logistic regression predicts a **probability** between 0 and 1 using the sigmoid (S-shaped) function.

II. MODEL EQUATIONS

Linear Regression assumes a direct linear relationship:

Logistic Regression models the **log-odds** of the event happening

III. MODEL ASSUMPTIONS

Linear Regression assumes:

- A linear relationship between variables

- Errors are normally distributed
- Constant variance of errors
- Observations are independent

Logistic Regression assumes:

- A linear relationship between predictors and log-odds
- Binary outcome variable
- No need for normally distributed errors
- Independent observations

IV. INTERPRETING COEFFICIENTS

Linear Regression

Each coefficient shows how much **Y changes** when X increases by one unit.

Example:

If $\beta = 5$, increasing age by one year increases the outcome by 5 units.

Logistic Regression

Coefficients represent changes in **log-odds**, not direct probability.

When exponentiated, they become **odds ratios**.

Example:

If $\beta = -0.05$, then each additional year reduces survival odds by about 5%.

V. HOW PARAMETERS ARE ESTIMATED

- **Linear Regression:** Uses Ordinary Least Squares (OLS) to minimize squared errors.
- **Logistic Regression:** Uses Maximum Likelihood Estimation (MLE) to find parameters that best explain the observed outcomes.

VI. MODEL EVALUATION METRICS

Linear Regression:

- R^2 and Adjusted R^2
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Logistic Regression:

- Confusion matrix
- Accuracy
- AUC-ROC
- AIC and BIC
- Pseudo R^2 measures

VII. WHY LINEAR REGRESSION IS BAD FOR BINARY DATA

Using linear regression for survival prediction can produce impossible values (like -0.2 or 1.5). It also violates statistical assumptions.

Logistic regression avoids this problem by **always producing valid probabilities between 0 and 1**.

VIII. TITANIC EXAMPLE

Incorrect (Linear Regression):

$$\text{Survival} = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Class})$$

This may predict unrealistic values.

Correct (Logistic Regression):

$$\log(p/(1-p)) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Class})$$

This outputs a probability such as **73% chance of survival**, which is meaningful.

IX. WHEN TO USE EACH MODEL

Use Linear Regression when:

- Outcome is continuous
- You want actual numerical predictions

Use Logistic Regression when:

- Outcome is binary
- You want probabilities
- You are solving classification problems

QUESTION 2.2: OVERALL SURVIVAL PROBABILITY

This value represents the **baseline survival chance** of a randomly selected passenger, without considering age, sex, or class.

It acts as:

- A reference point
- A null model
- A historical survival rate

QUESTION 2.3: SURVIVAL PROBABILITY ANALYSIS

I. BY PASSENGER CLASS

Passengers are grouped by class and survival rates are calculated.

Pattern observed:

- 1st class survived the most
- 3rd class survived the least

This reflects social status, cabin location, and access to lifeboats.

II. BY SEX

Females had a much higher survival rate than males due to the “**women and children first**” policy.

III. BY CLASS AND SEX

Combining both factors shows strong interaction effects:

- 1st-class females had the highest survival
- 3rd-class males had the lowest survival

IV. BY AGE GROUP

Children survived more often than older passengers due to evacuation priority and physical ability.

V. COMBINED ANALYSIS

When class, sex, and age are combined, survival chances vary drastically, showing that **multiple factors work together**.

VI. ROLE OF VISUALIZATION

Bar charts and grouped charts make these patterns easy to see and understand, especially interaction effects.

QUESTION 2.4: LOGISTIC REGRESSION MODEL MODEL SETUP

- **Response variable:** survived (0 = died, 1 = survived)
- **Predictors:** passenger class, sex, and age

The model estimates how these factors affect survival probability.

INTERPRETING RESULTS

- **Coefficients:** Show direction and strength of effects
- **p-values:** Indicate statistical significance
- **Odds ratios:** Explain how survival odds change

Example:

- Being male reduces survival odds
- Higher class improves survival odds
- Older age reduces survival odds

CONFIDENCE INTERVALS

They show the range where the true effect likely lies.

If the interval excludes 0 (or 1 for odds ratios), the effect is significant.

QUESTION 2.5: MODEL PERFORMANCE CONFUSION MATRIX

It compares predicted results with actual outcomes and helps identify:

- Correct predictions
- False alarms
- Missed cases

PERFORMANCE METRICS

- **Accuracy:** Overall correctness
- **Precision:** How reliable positive predictions are
- **Recall:** How many survivors were correctly identified
- **Specificity:** Correct identification of deaths
- **F1-score:** Balance between precision and recall

PRACTICAL MEANING

A good model:

- Explains historical survival patterns
- Confirms social and evacuation policies
- Helps understand risk factors

However, it remains **probabilistic**, not perfect, and cannot capture individual behavior.

QUESTION 3: PRINCIPAL COMPONENT ANALYSIS (PCA)

QUESTION 3.1: WHAT IS PRINCIPAL COMPONENT ANALYSIS (PCA)?

Principal Component Analysis (PCA) is a technique used in data mining and machine learning to **simplify complex datasets** that contain many variables.

Instead of working with many correlated variables, PCA transforms them into a **new set of uncorrelated variables** called **principal components**.

The main idea of PCA is to:

- Take many related variables
- Convert them into fewer independent components
- Keep as much important information as possible

These new components are ordered so that the **first component explains the most variation in the data**, the second explains the next most, and so on.

In short, PCA helps us **reduce dimensionality while preserving information**.

KEY PRINCIPLES OF PCA

A. Variance Maximization

PCA works by finding directions in the data where the **spread (variance)** is largest.

- **PC1** captures the largest amount of variance in the data
- **PC2** captures the largest remaining variance, but in a direction perpendicular to PC1
- Each next component captures as much remaining variance as possible
- This ensures that the most informative patterns appear first.
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B. Orthogonality

All principal components are:

- **Uncorrelated**
- **Perpendicular to each other** in geometric space

This means each component carries **unique information** and removes the problem of multicollinearity that often affects regression and machine learning models.

C. Linear Transformation

PCA creates **linear combinations** of the original variables.

The weights (w) come from eigenvectors and determine how much each original variable contributes to a component.

PCA therefore **preserves linear relationships** in the data.

D. Dimensionality Reduction

Originally, the dataset has **p variables**.

After PCA, we keep only **k components**, where $k < p$.

- High-variance components are kept
- Low-variance components (mostly noise) are removed

This results in a simpler dataset that still represents the original structure well.

APPLICATIONS OF PCA IN MACHINE LEARNING

A. Feature Reduction

When datasets have hundreds or thousands of features, PCA helps by:

- Reducing computation time

- Lowering memory usage
- Avoiding the curse of dimensionality

Examples include image data, genetic data, and text analysis.

B. Data Visualization

High-dimensional data cannot be visualized easily.

PCA allows us to project data into **2D or 3D**, making it possible to:

- See clusters
- Identify patterns
- Detect outliers

This is useful in customer analysis, finance, and biology.

C. Noise Reduction

PCA assumes that:

- High variance represents meaningful signal
- Low variance often represents noise

By removing low-variance components, PCA helps clean noisy data such as images and sensor readings.

D. Handling Multicollinearity

When predictors are highly correlated, PCA converts them into **uncorrelated components**, leading to:

- More stable models
- Better regression performance
- Reliable coefficient estimates

E. Data Compression

PCA reduces storage and transmission needs by:

- Storing fewer variables
- Allowing approximate reconstruction of original data

This is widely used in image and video compression.

F. Anomaly Detection

Normal data fits well within the PCA space.

Outliers show **large reconstruction errors**, making PCA useful for:

- Fraud detection
- Quality control
- Network security

WHY PCA IS USED ON EXPLANATORY VARIABLES

Curse of Dimensionality

Too many features reduce model performance. PCA reduces dimensions and improves generalization.

Multicollinearity

Highly correlated variables inflate variance. PCA removes correlation completely.

Computational Efficiency

Fewer dimensions mean faster algorithms and lower memory usage.

Visualization

PCA allows complex data to be seen and understood visually.

Noise Filtering

Removing low-variance components improves signal-to-noise ratio.

Feature Engineering

PCA creates new, informative features without relying on domain-specific rules.

LIMITATIONS OF PCA

- Captures only **linear** relationships
- Principal components are harder to interpret
- Sensitive to variable scaling (standardization is required)
- High variance does not always mean high importance
- Some information is always lost during reduction

QUESTION 3.2: MATHEMATICAL FOUNDATION OF PCA

PCA begins with a **data matrix X**, where:

- Rows represent observations (e.g., trading days)
- Columns represent variables (e.g., stocks)

The data is first **standardized** so that all variables have mean 0 and variance 1.

Next, a **covariance or correlation matrix** is computed to understand relationships between variables.

Eigenvalues and eigenvectors are then extracted:

- **Eigenvalues** represent how much variance each component explains
- **Eigenvectors** define the direction of each component

The original data is finally transformed into **principal component scores**, which are uncorrelated and ordered by importance.

QUESTION 3.3: DOW JONES STOCK PCA ANALYSIS

DATA DESCRIPTION

- 30 Dow Jones stocks (with AMZN replacing WBA)
- Daily returns from January 2020 to January 2021
- Approximately 252 trading days

2020 MARKET CONTEXT

The year 2020 was dominated by:

- COVID-19 pandemic
- Extreme market volatility
- Technology sector dominance
- Policy interventions

This caused **high correlation across stocks**, making PCA especially useful.

INTERPRETING PC1 AND PC2

PC1:

- All stocks have positive weights
- Represents overall market movement
- Acts like a market index
- Explains the largest share of variance

Conclusion: PC1 is effectively the **market factor**.

PC2:

- Mix of positive and negative weights
- Represents differences between sectors or styles
- Orthogonal to PC1

QUESTION 3.4: VARIANCE EXPLAINED AND SCREE PLOT

A variance table shows how much information each component explains.

Typically:

- PC1 explains the most
- Variance drops sharply after a few components

A **scree plot** helps identify the point where adding more components gives little benefit (the “elbow”).

Using a **95% cumulative variance rule**, PCA often reduces 30 stock variables to about **8–12 components**, preserving most information while greatly simplifying the data.

QUESTION 3.5: SCATTER PLOT AND OUTLIER ANALYSIS

A scatter plot of PC1 vs PC2 shows how stocks are positioned in factor space.

Stocks far from the average position are considered **unusual**, meaning:

- They behave differently from most stocks
- They have unique risk or sector exposures

Distance from the center is measured using **Euclidean distance**.

WHY SOME STOCKS ARE UNUSUAL

- High PC1 → strong market sensitivity
- Low PC1 → defensive behavior
- Positive or negative PC2 → sector or style differences

COVID-19 intensified these patterns, especially for:

- Technology stocks
- Defensive healthcare stocks
- Energy and travel-related stocks

PRACTICAL IMPLICATIONS

For Portfolio Management

- Unusual stocks improve diversification
- Defensive stocks reduce risk
- Growth stocks increase upside potential

For Risk Management

- PC1 measures market risk
- PC2 measures sector/style risk
- Distance measures idiosyncratic risk

For Investment Strategy

- PCA supports factor-based investing
- Enables market-neutral strategies
- Helps identify alpha opportunities